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PNG 516 — Past Questions & Materials
by Renance

Course: PNG 516
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — 2017 PQ (PNG 516/PEE 515)

Level 500 • First Semester • PastQuestion

PNG 516

Mat. No. _____ Mid-Semester Quiz

1. Define natural gas reservoir fluid processing.

Fluid mixture of hydrocarbon, water, solids, inert gas and other impurities found in porous permeable rocks called Separation natural gas reservoir fluid into its component phases and the treatment of the phases to their respective sales or disposal qualities

2. Practically speaking, propane and butane, commonly referred to as LP are liquid under Nigerian normal pressure and temperature, hence referred to as extremely volatile liquids

a. TRUE

b. FALSE

3. In the table below, tick(✓) reasons why the followings are removed from natural gas during processing

No Compound Corrosi Toxi Heating Econo Increa Forms Aci Sluggi

. d ve c value mic se solid on d ng

reductio value flow cryogenic Rai Flow

n volum Refrigerati n

e on

1 HC Liquid ☐ ☐ ☐ ☐ ☐

2 H₂S ☐ ☐ ☐ ☐ ☐

3 Nitrogen ☐ ☐ ☐

4 Water ☐ ☐ ☐ ☐ ☐

5 CO₂ ☐ ☐ ☐ ☐ ☐

6 Rare ☐ ☐ 25

Gases

4. Identify the phase diagrams in the figure below with respect to types of hydrocarbon

A: _____ Gas _____

B:

_____ Condensate _____

C: _____ Light

Crude _____

D: _____ Heavy

Crude _____

5. In processing/treating natural gas for sale, why does sweetening precede water and hydrocarbon liquid dew-pointing?

So that proceeding products and processes are not

contaminated

6. Identify the processes/equipment natural gas from both oil well and natural gas well undergoes with the respective products as indicated by the figure below

PNG 516

Mat. No. _____ Mid-Semester Quiz

A: __ Reservoir fluid _____

B: __ Expander(JT-Valve) _____

C: __ Separator _____

D: __ LP Gas _____

E: ---- Condensate _____

G: __ Cooler _____

H: __ Compressed AG HP Vapor Gas __

I: __ Expanded NAG HP Gas _____

J: __ Sales Natural Gas _____

7. Define Natural Gas Utilization

Putting natural gas into economic and environment use.

8. Mention two main refrigeration systems used in processing of natural gas and three(3) main processes they are used for

Refrigeration Systems Main Processes in Natural Gas

Processing

1.Mechanical Compression 1.Natural Gas Hydrocarbon dew-
Refrigeration System pointing

2. Natural Gas Water dew-pointing

2.Expansion refrigeration 3. LNG Production
system

9. Mention three main features of temperature reduction achieved by Joule-Thompson effect on gas stream expansion

a. __ No work done _____

b. __ No work taken _____

c. __ Constant Enthalpy _____

10.In a sequential order, mention the equipment/devices on the path of the information/signal from the sensor/measuring device to the operator at the main control room of the SCADA system

Sensor - Remote terminal unit(RTU) - Satellite broadcast system -

Satellite - Satellite reception system - Human Man Interface (HMI) -

Operator

11.Apart from the pipes, mention six(6) main facilities or equipment you must have in natural gas distribution network to get the gas to the customer

a. __ Compressor _____

b. __ Pressure regulator _____

- c. ____ Meter _____
- d. ____ Storage facility _____
- e. ____ Injection station _____
- f. ____ LNG facility _____

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Mat. No. _____ Mid-Semester Quiz

12. Mention three main storage facilities for sale natural gas

- a. ____ Underground storage _____
- b. ____ Metal fabricated tanks _____
- c. ____ Insulated Fabricated tanks _____

13. Mention three(3) main benefits of depleted reservoir storage facility

- a. ____ Facility already exist and no further expense on new construction
- b. ____ High porous and permeable rock for storage _____
- c. ____ Cheapest and easiest to operate _____

14. In this type of fluid flow, the fluid molecule follows in a wavy manner but in parallel path that is not parallel to the boundaries of the fluid conduit

- a. Turbulent flow
- b. Transition flow
- c. Laminar flow
- d. None of the above

15. What is the flow type of a 0.78 specific gravity natural gas flowing at 5.8 mm scf/day in a 12 ¼" ID tubing if the viscosity of the gas is 4.75 cp?

= = 1,562.75 $\square$ 20000

$\square$ Fluid flow is Lamina

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Course: PNG 516 - Natural Gas Engineering II Mat. No. _____

Date: Monday 20th March, 2017 Time: 11.00 am - 2.00 pm

Basic Instructions:

1. Answer question no. 1 and four(4) others
2. Some basic equations and constants are provided below

20,100 Q g $\square$ g C $\square$

1. N Re = 2. qm = . $\square$. . d 2 . 2 $\square$ P. $\square$ 1

D $\square$ 1 – $\square$ 4 4

3. n 4.

5. 1 psi = 27.707 in of water 6.

T $\square$ P 2 – P22 $\square$

$\square$ 2 2 $\square$

$$0.5 q_g = 433.49 b_d 2.667 \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$2.667 \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$\left(\frac{L}{Z T} \right)$$

$$T$$

$$7. q_g = 433.49 b_d$$

$$\left(\frac{L}{Z T} \right) 8. P_b$$

$$P_b \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$T C \left(\frac{1}{h} \right) P$$

$$10. q_h = 218.59 b_d \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$9. W = D. M \times 100 P_b \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$\left(\frac{1}{g} \right) T. Z$$

$$K_i + D. M \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$11. 12. \left(\frac{1}{P_1} - \frac{1}{P_2} \right)$$

$$T = 13.47 \ln(P) + 34.27 \ln \left(\frac{1}{g} \right) - 1.675 \ln(P) \left(\frac{1}{g} \right) - 20.35 \left(\frac{1}{g} \right)$$

$$1$$

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1.

The Nigerian-Algerian Transcontinental pipeline with flow capacity of 250 mm.scf/day operating at 900 psia and 90 0F, transmits Utorogu natural gas with composition given below. The pipeline pressure is expected to be maintained with compression stations along the route to Spain whose average winter temperature is 38 0F.

- What is the maximum lb. water/mm.scf this gas can hold at the operating pressure and temperature? At 900 psia and 90 0F, water content = 50 lb.H₂O/mm.scf
- Employ Katz equation to determine the hydrate formation temperature at the operating pressure.
- If this transmission requires further dehydration at the Utorogun gas plant, how many gallons of water should be removed per day considering Spain temperature conditions.
- If the alternative of hydrate inhibition is taken, what will be the TEG (MW= 150.174 g/mol) glycol injection rate in pounds per day to prevent hydrate formation? - $K_i = 2220(0C) \text{ or } 4000(0F)$

Component Mole % MW ZiMW

N 8.1 28 2.268

C1 76.7 16 12.272 5

C2 7.1 30 2.13

C3 3.5 44 1. 54

i-C4 1.7 58 0.986

n-C4 1.1 58 0.638

C5+ 1.8 86 1.548

100.0 21.382

Gas Specific Gravity = $21.382/28.9625 = 0.738$

□ □

$$\begin{aligned}
 T &= 13.47 \ln(P) + 34.27 \ln(\gamma_g) - 1.675 \ln(P) \ln(\gamma_g) - 20.35 \\
 &= 13.47 \ln(900) + 34.27 \ln(0.738) - 1.675 \ln(900) \ln(0.738) - 20.35 \\
 &= 13.47(6.8) + 34.27(-0.304) - 1.675 [6.8 (-0.304)] - 20.35 \\
 &= 91.596 - 10.42 - 10.88 - 20.35 = 49.945 \text{ } ^\circ\text{F}
 \end{aligned}$$

1

At 38 °F and 900 psia, natural gas max. water content = 9 lb.H₂O/mm.scf

Taking 35 °F and 900 psia, to be on safer side natural gas max. water content = 8.5 lb.H₂O/mm.scf

□ 50 - 8.5 = 41.5 lb.H₂O/mm.scf x 250 mm.scf/day = 1245.5 gal/day will be removed

8.33 lb/gal

For Inhibition: D = 90 - 35 = 55; MW 150.174 g/mol; K_i = 4000

$$W = D \cdot M \times 100 = 55 \times 150.174 \times 100$$

$$K_i + D \cdot M \quad 4000 + 55 \times 150.174$$

$$= 67.372\%$$

1

$$\text{Glycol injection} = 0.67372 \times 41.5 \times 250 = 6,989.845 \text{ lb/day}$$

2

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2.

a. Name the four(4) basic elements of the Offshore Field Development, give an example of an offshore transportation system that is related to each element and state the function of the transportation system to the respective Offshore Field Development element

i. Fixed/Floating structures

ii. Subsea system

iii. Flowline/Pipeline/Riser system

iv. Topsides processing system

b. What is a Jumper in the offshore pipeline system?

1

i. Jumper: Short flowline connecting a subsea well back to its manifold

c. Give a similarity and a difference between Transport Line System and Export Transport Pipeline System

No. Differences Export Transport Transport Similarities

Pipeline System Pipeline System

1 Function Transport fluid from Transport the Transport fluids

a field to the shore, fluid from

export terminal different fields

(onshore or to a treatment

8

offshore) or to a or loading

manifold for joining facility

a transport line

2 Pipe Diameter medium diameter Generally large Monophasic

(up to 24''), diameter (20'' flow

to 40'')

3 Length medium to long Medium length Pipeline System

length (several

hundreds km)

Hook-up Joins a transport Does not join

line at Manifold Export line

d. Name and briefly describe three(3) main processes to be incorporated into your pipeline design to mitigate against corrosion and pipeline environmental degradation

i. Material Selection/Design

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a. Corrosion Resistant Alloy Steel - nickel and/or chromium

b. If fluid to be transported is too corrosive for carbon steel its resistance is improved by addition, lining and /or cladding it with nickel and/or chromium

c. Main challenge is that it cost about 4 times more and welding process is slow and more complex

d. Examples are:

i. 13% Chromium Stainless Steel

ii. Duplex Stainless Steel(2205)

iii. Duplex Stainless Steel(2507)

3

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ii. Cathodic protection

a. Internal Corrosion Protection

i. Add provisional steel material – i.e. give corrosion allowance

ii. Use Internal Coating – flow coating

iii. Inject corrosion inhibitors into the production

iv. Use CRA

b. External Corrosion

i. Anti-corrosion coating - Primary protection

1. Corrosion Coatings

Insulation Coatings

Concrete Weight Coating

Field Joint Coating

2. Thermoplastic
3. Elastomers
4. Syntactic foams
5. Compacted microspheres
6. Aerogels

iii. Cathodic Protection - Secondary protection

iv. Corrosion Inhibitors

The use of corrosion inhibitors has proven to be the effective method to mitigate internal corrosion in wet sour gas pipelines

3.

a. Give the definitions, functions and areas of application of the facilities listed below in natural gas transmission

i. Chemical injection

1. Injection of pungent smell gasses or designated chemicals such as mercaptans into the gas stream to achieve the objective
2. It functions by providing very irritating pungent smell for ease of identification of the natural gas in the of leakage 3
3. Mainly at the main gate station of a distribution network

ii. Pressure regulation

1. Pressure regulation is normally the reduction of natural gas pipeline pressure to the level of the next service area of the pipeline network 3
2. It functions to bring the pipeline pressure to the level of customer usage
3. At main gate station, entrance of area of the distribution network and the entrance of service line

iii. Metering

1. Measurement of natural gas volume and/or energy for custody transfer
2. Functions for gas seller and customer to know accurate amount of 3 volume or energy transferred, for record and billing purposes
3. At main gate station, entrance of area of the distribution network and the entrance of service line

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b. Give a well labelled block sketch of a typical combined chemical injection, pressure regulation and metering station at the main gate of a city natural gas distribution pipeline system.

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c. With a well labelled block diagram, describe the process of natural gas compression in a typical natural gas transmission truck line compression station

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4.

a. State the facilities/equipment that found in a typical natural gas distribution pipeline network system.

i. The Pipe

The Valves

Other Monitoring, Measurement, instruments, etc. equipment attached to the pipe

Scratchers (Pigging) Station

Compressor stations 5

Pump stations

Metering stations

Regulator stations

Delivery stations

Holders

Fabricated assemblies.

b. In a tabular format, list the main pipeline types with their respective functions and facilities/equipment in each pipeline type.

No. Main Pipeline Types Facilities/Equipment

Gathering Pipeline Network Flow line

Compression station

Flow line valves

Manifold

Control and Monitoring system

Trunk line Pipeline Larger diameter pipe

Compression station

Block valve station 6

Manifold

Metering/Pressure Regulation Station

Pigging stations

Control and Monitoring system

Distribution pipe line Various sizes of pipe diameter

Metering/Pressure Regulation Station

Chemical Injection station

Storage facility

LNG facility

Compression station

Control and Monitoring system

c. With the pipeline types and facilities/equipment in (b) above, give a conceptual design block diagram of

natural gas distribution pipeline network with the following

- i. LNG plant,
- ii. Underground storage facility
- iii. Cement factory
- iv. Brewery
- v. 3(nos.) residential estates
- 3
- vi. 5(nos.) isolated bungalows
- vii. A Church
- viii. A Musk
- ix. A town hall.

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8

a.

Desmond Res. Area Res. Area

Steel plant B A

(200 psia) (100 psia) (150 psia)

2.5 km

0.2 km 0.5 km

80 mm.scf/d A

50 mm.scf/d 60 mm.scf/d

From processing plant TRUNK LINE Main

B

25 km; 400 mm.scf/d; Gate

3.4 km, 450 psia

1000 pia (850 psia)

200 mm.scf/d

1.5 km

C

40 mm.scf/d

2 km

150

mm.scf/d Comm.

Res. Area

Flippant (250 psia)

Glass Factory

(150 psia)

5.

a. Applying the principles of Bernoulli equation of conservation of energy and basic assumptions of continuity and incompressibility, use appropriate diagrams and the interrelationship between the pressure drop, flow velocity, mass flow rate and flow area across a constriction to develop the basic flow equation for the orifice plate flow capacity

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

2

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

a. Applying the principle of Continuity

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

b. Now applying the Assumption of Fluid being incompressible

$$\rho_1 = \rho_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

Assumption of flow continuity means

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

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Recall that

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

d. Bernoulli's Law describing the behaviour of an ideal fluid in a closed system and states that the fluid overall energy as it enters and leaves the system remain the same – conservation of energy.

i. Applying this means that as the potential energy is converted to kinetic energy as the fluid speeds up and the pressure drops at the nozzle throat,

Total change in the static pressure = total change in dynamic pressure, $\Delta P_S = \Delta P_D$

ρ_2

$\rho_2 u_2 = \rho_1 u_1$

$\rho_2 u_2^2$

ii. And since $P = P_D = P_u$, $\rho_2 P =$

$\rho_2 u_2^2$

iii. Substituting u_1 and u_2 from continuity in (ii) above gave the orifice flow equation

$\rho_1 u_1^2$

$q_m = \rho_2 \cdot d^4 \cdot 2 \cdot \rho_1 \cdot P$

$\rho_1 d^4$

4

$\rho_1^2 - 4 \rho_1$

$\rho_1 D^4$

b. Use the orifice plate flow capacity equation in field units to calculate the capacity for the following flow condition of a typical orifice plate installation on a 4½ " ID pipe

$T_b = 65^\circ\text{F}$; $T_o = 118^\circ\text{F}$; $P_b = 15\text{ psia}$; $P_o = 800\text{ psia}$ (pipeline operating pressure)

$\rho_P = 98\text{ psi}$; $Z = 0.812$; $\gamma_g = 0.78$; $C = 0.678$; $\beta = 0.77$; $\beta = 0.56$

$T_b = 65 + 460 = 525^\circ\text{R}$; $T_o = 118 + 460 = 578^\circ\text{R}$

$1\text{ psi} = 27.707\text{ in of water}$ $\rho_2 = 98 \times 27.7 = 2,714.6\text{ lb/ft}^3$

$T_C = \rho_2 \cdot P \cdot 525 \cdot 0.678 \cdot 2,714.6 \times 800$

$q_h = 218.59 \cdot \beta \cdot d^2 \cdot \rho_2 = 218.59 \cdot 0.77 \times 4.5^2$

$P_b \cdot 1 - \beta^4$

$\gamma_g \cdot T \cdot Z \cdot 15 \cdot 1 - 0.56^4$

$0.78 \times 578 \times 0.812$

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6.

a. In the table below, tick(☐) reasons why the followings are removed from natural gas during processing

No. Compound Corrosive Toxic Heating Economic Increase Forms solid Acid Slugging

value value flow on cryogenic Rain Flow

reduction volume Refrigeration

1 HC Liquid ☐ ☐ ☐ ☐

2 H₂S ☐ ☐ ☐ ☐

3 Nitrogen ☐ ☐

4 Water ☐ ☐ ☐ ☐

5 CO₂ ☐ ☐ ☐ ☐

6 Rare Gases ☐ ☐

b. Define natural gas utilization

Natural Gas utilization involves putting natural gas into economic and environment use.

c. With four(4) main areas of interest chosen, briefly describe the edge of the natural gas over coal, oil and crude oil products as fuel

No. Area Natural Gas

1 NG Power Plants 1. More Efficient

2. Require less fuel input for same amount of power generated

3. Emit less Co2

4. Cheaper

5. Quicker to start-up and shut down

Transport 1. Cleaner

2. Less Fuel Usage

3. More Efficient 7

4. Less NOx

5. Less Fine Dust

6. Less CO2

Environmental Less Acid gas

Less greenhouse gas

Less CO2

Less particulate materials

Burns clean

Cooking Cheaper

Burns cleaner

Produces more heat energy

Utilization More worldwide as fuel

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d. Give four(4) benefits of underground salt dome storage facility

i. Provides Larger storage capacity

ii. Cheaper to operate

iii. Easy to operate 2

iv. Surface can be used

v. Provides high facility security

vi. Provides impermeable and self-sealing storage and therefore less room for leakage

vii. Less facility maintenance

e. Briefly describe with appropriate diagrams the filling and production of underground salt dome storage facility.

Construction

A well is drilled about a 1000 ft into the salt dome

Two(2) casings; product casing and displacement casing

2

On the surface, a brine pond is constructed and appropriate well head with valves are installed to inject in and withdraw out product through the casings annulus and to inject in and withdraw out brine through the displacement casing.

Portable water is injected in to dissolve out the salt in form of brine thereby creating the storage volume

3

Storage Facility Operation 1

Filling

Product is injected in through the casings annulus which displaces the brine out through the displacement casing to the surface and into the brine pond.

Well head valves are shut in at the end of filling.

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Production

During production, brine is pumped into the salt dome from the brine pond through the displacement casing

1

Product is displaced out to the surface through the annulus of the two casings

4

50

9

38 90 0

F

11